RHEA NAIR

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EDUCATION

University of Colorado Boulder — *Expected May 2027*
M.S. in Computer Science

Virginia Polytechnic Institute and State University (Virginia Tech) — *May 2025*
Dual Degree: M.S. in Business Analytics (MSBA) — *CGPA: **3.97/4.00***
and B.S. in Cybersecurity Management & Analytics

Narsee Monjee Institute of Management Studies (MPSTME) — *May 2023*
B.Tech in Computer Science (Data Science) — *CGPA: **3.70/4.00***

SKILLS

Languages	Python, Java, C++, MATLAB, R, SQL, NoSQL
Machine Learning & AI	PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, CNNs, Transformers, Deep Learning, LLM Architectures, AI/ML Integrations
Data & Libraries	Pandas, OpenCV, MongoDB, Hadoop, Apache Spark, Snowflake, Data Modeling Frameworks
Systems & Tools	Git, Linux, Docker, AWS, Azure, DigitalOcean, Databricks, Tableau, PowerBI, Excel, SAS, Streamlit, Dash, A/B Testing, CUDA, Causal Inference, BI Tools, Information Security

EXPERIENCE

Full Stack Data Analytics Engineering Intern — *Jan 2026 – Present*
Ascend Cargo Systems

- Led the integration of an ***agentic AI chatbot*** into ***Apache Superset***, enabling self-service BI that allows clients to autonomously query data via natural language, accelerating time-to-insight and reducing ad-hoc reporting bottlenecks.
- Built the company's first ***data warehouse and ETL pipelines*** over a ***71-table production MySQL schema***, creating a clean, structured data foundation for downstream analytics and machine learning applications.
- Developed ***analytical dashboards*** and complex ***SQL reporting queries*** in ***Apache Superset*** surfacing KPIs such as tender acceptance rate and late shipment aging to a client.
- Deployed an ***embedded BI environment*** on ***Kubernetes*** with role-based access and Row Level Security, ensuring governed, client-specific data visibility across the production platform.

Research Assistant — *May 2024 – Jul 2024*
Virginia Tech, Blacksburg, VA

- Reduced the research paper review cycle from 14 days to 2 days by developing ***Python scripts*** with ***Pandas and regex-based automation***, enabling faster parsing, metadata extraction, and error detection, which improved workflow efficiency by **85%**.
- Applied ***NLP classification models*** using ***scikit-learn and spaCy*** to automatically categorize academic documents, reducing manual review time by 70% and improving classification accuracy across large datasets.
- Automated ***data collection and preprocessing pipelines*** with ***BeautifulSoup, Pandas, and NLTK***, creating scalable workflows that ensured reproducibility and standardized processing for faculty research.
- Created ***Python-based summary reports and visualizations*** with ***Matplotlib and Seaborn*** that highlighted document trends, provided actionable insights, and supported faster, data-driven faculty decision-making.

Data Scientist and Researcher — *Jan 2023 – Jan 2024*
Tata Institute of Fundamental Research, Mumbai, India

- Designed and executed ***multi-agent reinforcement learning simulations*** in ***Python (PyTorch)***, managing large-scale experimental data stored in both ***NoSQL (JSON)*** and ***relational SQL*** formats to optimize access and processing.
- Engineered a ***Neuro Evolution-based deep RL model*** using ***PyTorch and sklearn*** , improving training efficiency by **85%** compared to baseline methods and reducing computational resource consumption.
- Leveraged ***Hadoop and Apache Spark*** for distributed data preprocessing and analysis, ensuring scalability and reducing computation time across simulation workloads.
- Authored comprehensive ***technical documentation and experiment reports***, including methodology breakdowns, performance comparisons, and workflow diagrams, to support collaboration and internal knowledge transfer.

PROJECTS

BoulderMove (Python, FastAPI, XGBoost, GTFS, OSMnx, Docker, Google Cloud) [Link] — *Aug 2025 – Nov 2025*

- Built a ***multimodal trip planning platform*** integrating ***GTFS transit data*** and ***OSMnx routing*** for optimized travel paths across transit and pedestrian networks
- Developed an ***XGBoost prediction model*** to estimate transit arrival reliability using weather and congestion features.
- Deployed scalable APIs with ***FastAPI*** and ***Docker*** on ***Google Cloud*** for real-time route optimization.

Brain Tumor Detection using CNNs (Python, TensorFlow, Keras, OpenCV) [Link] — *Apr 2023 – Jul 2023*

- Built an end-to-end ***Convolutional Neural Network*** to classify brain MRI scans as tumor vs. non-tumor, trained from scratch in ***TensorFlow/Keras*** and achieving **0.92 validation accuracy** with stable precision and recall across classes.
- Designed a ***custom augmentation pipeline*** (rotation, flips, zoom, brightness/contrast) that expanded the dataset from **253** to **2065** images, improving class balance, generalization, and robustness.
- Implemented full ***preprocessing & experiment tracking*** pipeline using OpenCV-based brain-region cropping, 240×240 resizing, normalization, TensorBoard logs, and saved checkpoints for reproducibility.